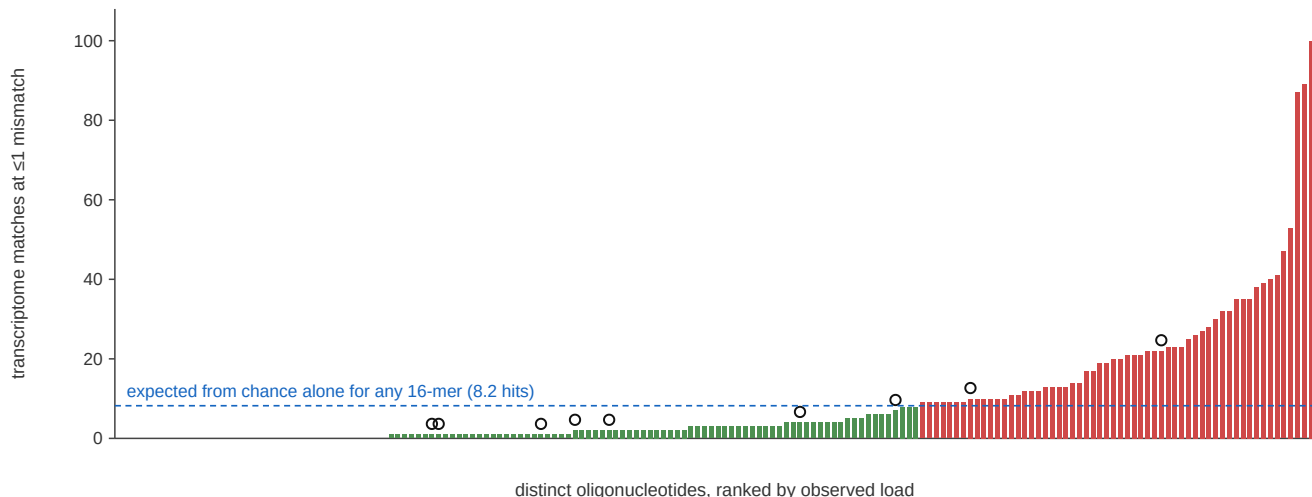


Transcriptome load per distinct oligonucleotide against chance expectation

176 distinct oligonucleotides at real exon junctions, ranked; exact plus ≤ 1 -mismatch matches over 186,185 transcripts.

○ the 9 designs spanning two or three partners' seams, each one molecule plotted once.



One bar is one molecule and not one design record: the 190 design records at real exon junctions are 176 distinct oligonucleotides, because the 9 marked designs each span more than one partner's seam and enter the series once rather than once per seam.

The chance expectation assumes independent uniform bases; the transcriptome span it is computed over is the exhaustive scan's measured one. It separates "more than chance" from "at chance" and is not a significance test. 118 of 176 bars fall at or below that line.

Counts are predictions from sequence search, not measured off-target activity. Bars are green at or below the line and red above it, which repeats each bar's own height and carries no information of its own. 58 of the 176 plotted designs sit above it; on the mean the set comes in at chance — 9.2 observed against 8.2 expected, a ratio of 1.12, while the median is 3, so the excess is a long right tail rather than a shift of the whole set.

40 of the 176 bars are zero and therefore draw nothing at all, so 22.7% of the abscissa is blank. The series begins at the left edge, not where the first visible bar is.

Not plotted: 10 designs at 2 modelled breakpoints rather than real exon junctions; 4 of those sit above it, at 16–95 matches and 75.0–81.2% GC.